

EVoke – Business Requirements Document (BRD): Location Module

1. Purpose & Scope

The **Location Module** in EVoke is developed to enable **Charge Network Operators (CNOs)** and their associated **Charge Station Operators (CSOs)** to efficiently create, manage, and configure physical charging sites. Each **location** represents a real-world EV charging site under a CSO-type organization and serves as the foundation for charge station deployment and pricing configuration.

This module ensures that every site in the EV network is properly defined with organizational, financial, and operational context — forming a key part of EVoke's infrastructure management hierarchy.

2. Relationship with Organization Module

The **Organization Module** forms the top level in the EVoke hierarchy, where various organization types are created — **CSO**, **CSR**, **FSO**, and **TPO**.

- Only **CSO-type organizations** can create locations under them.
- Each location inherits the association of its parent CSO and linked reseller (CSR).
- The **Location Module** sits directly below the Organization Module and above the **Charge Station Module**.

Hierarchy:

Organization → Location → Charge Station

This hierarchical relationship ensures smooth data flow, accountability, and reporting across operational, financial, and technical levels.

3. Stakeholders

- **Charge Network Operator (CNO):** Oversees network-wide infrastructure, policies, and financial settlements.
- **Charge Station Operator (CSO):** Owns and operates the physical sites (locations) where chargers are installed.
- **Charging Service Reseller (CSR):** Linked to CSOs for revenue sharing, billing, and reporting.
- **End-Users (Drivers):** Indirect stakeholders who benefit from well-managed locations, transparent pricing, and reliable charging availability.

4. Functional Overview

The **Locations Module** provides the tools to define, configure, and manage charging sites within the EVOKE ecosystem. It ensures that all site-level parameters — including address, utility data, network fees, and operating hours — are centralized and easily accessible.

Key capabilities include:

- Adding new locations under CSO-type organizations.
- Linking resellers and utilities.
- Defining address and geographical coordinates.
- Setting up CNO and CSR fee structures.
- Managing payout accounts and site contacts.
- Configuring operating hours for site accessibility.
- Viewing and managing existing locations through listing and map-based views.

5. Functional Requirements

5.1 Location Listing & Dashboard

The **Location Dashboard** displays all created locations with key operational and financial details.

Displayed Columns:

- Actions
- Organization
- Location
- Address
- Business Category
- Reseller
- Utility Name
- Coordination Node ID
- Sales Tax
- CNO Fee Percent
- CSR Fee Percent
- CNO Session Fee
- CSR Session Fee

Dashboard Features:

- **Filter Panel:** Filter by Organization, Location, Reseller, Utility Name, or Coordination Node ID.

- **Export to Excel:** Download complete list of location details for analysis or reporting.
- **Show/Hide Columns:** Customize visible columns.
- **Refresh Icon:** Reload latest dashboard data.
- **Map View:** Visualize site locations geographically alongside tabular data.

5.2 Add Location Workflow

Adding a new location involves three key segments.

Segment 1 – Location Details

Fields:

- **Organization:** Drop-down; only **CSO-type organizations** appear.
- **Location Name:** Text field; mandatory (3–30 characters).
- **Business Category:** Drop-down; optional; data maintained via backend.
- **Reseller:** Drop-down; optional; displays **CSR-type organizations**.

After saving this segment, the system moves to the **Address Segment**.

Segment 2 – Address

An interactive **map interface** allows address entry with real-time search and auto-population.

Editable:

- Address Line 1 (*mandatory*)
- Address Line 2 (*optional*)

Auto-Populated (non-editable):

- Latitude
- Longitude
- City
- State
- Country
- Zip

Users select **Save and Continue** to proceed to utility details.

Segment 3 – Utility Details

Defines utility and energy supplier information critical for charging operations.

Fields:

- **Utility Name:** Drop-down (Mandatory; fetched from Utility Module).
- **Utility Cost:** Mandatory; allows decimal values up to **three decimal places**.
- **Meter ID:** Optional.
- **Coordination Node ID:** Optional.

After saving, the new location appears at the **top of the listing page** for quick access.

6. Location Actions

Each location supports multiple contextual actions under the **Actions column**.

6.1 Edit Location

Allows users to edit any field across the three segments.

Validations from the creation flow (e.g., field limits, mandatory rules) apply here as well.

6.2 Network Fee Configuration

Purpose

To define transparent financial relationships among CNO, CSR, and CSO entities for each location.

Fee Configuration During Location Setup

- **CNO Fees:**
 - *Session Fee (Fixed per session):* Default \$0.50.
 - *Fee % (Percentage of transaction total):* Default 7%.
- **CSR Fees:**
 - *Session Fee (Fixed per session):* Customizable.
 - *Fee % (Percentage of transaction total):* Customizable.

This ensures each site's financial terms are defined and traceable at setup.

6.3 Pricing Policy Integration

Objective

To establish how session fees and percentages defined at the Location level influence downstream **Pricing Policies** and **Settlement Calculations**.

Linkage Rules

- A **Pricing Policy** must always be tied to a **specific Location**.
- If a Pricing Policy is created directly from a **Charge Station**, the Organization and Location fields are auto-filled.
- Lists are filtered to prevent cross-location misconfiguration.

Fee Calculation Logic

- **Total Session Fee = CNO Transaction Fee + CSO Session Fee**
- **Transaction Total = Session Fee + Energy Fee + Idle Fee + Tax**
- **CNO Fee = CNO Session Fee + (CNO Fee % × Transaction Total)**
- **CSR Fee = CSR Session Fee + (CSR Fee % × Transaction Total)**
- **Net CSO Total = Transaction Total – (CNO Fee + CSR Fee)**

6.4 View Location

Displays full, read-only details of the selected location, including address, reseller, utility, and network fee information.

6.5 Attach Payout Account

Links a payout account (from Organization module) to the location for managing financial settlements.

Multiple payout accounts can be linked and managed per site.

6.6 Site Contacts

- Displays or allows the addition of **site-level contacts**.
- Auto-populates "Location - Site Admin" contacts created during Charge Station setup.
- Fields include:
 - First Name, Last Name, Contact Type, Contact Number, and Email.
- Contacts can be **added or deleted** directly from this interface.

6.7 Operating Hours

Defines working hours for the location.

Fields:

- **Select Week Days:** Drop-down (Select All, Monday–Saturday).
- **Period Begin / End:** Time pickers (12:00 AM–11:30 PM, 30-min increments).

Validation Rule:

"Period End should be after Period Begin."

Users can define multiple time slots or delete existing ones to adjust the operational window.

7. Non-Functional Requirements

- **Security:** All organizational, location, and financial data must be securely handled with appropriate user access controls.
 - **Auditability:** Each action (create, update, delete) is logged for compliance and traceability.
 - **Scalability:** Designed to handle numerous organizations, locations, and stations without performance degradation.
 - **Performance:** Optimized address search, map rendering, and filtering for minimal latency.
 - **Validation:** Robust front-end and back-end validation for input accuracy and data consistency.
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8. Summary

The **Location Module** establishes a structured, transparent, and flexible framework for defining and managing charging sites within the EVoke ecosystem.

By combining **operational detail**, **financial configuration**, and **geographical precision**, it ensures every charging site under a CSO organization is effectively managed, financially accountable, and seamlessly integrated with downstream modules like Pricing, Charge Stations, and Subscriptions.

9. Glossary of Key Terms

Term	Definition
CNO (Charge Network Operator)	The entity owning and overseeing the EV network infrastructure, responsible for managing subscriptions, fees, and settlements.

CSO (Charge Station Operator)	The operator responsible for managing daily operations of charging sites, chargers, and customer-facing services.
CSR (Charging Service Reseller)	A business partner linked to CSOs for billing and financial settlement activities.
FSO (Field Service Organization)	Provides maintenance and operational support for EV stations. Currently not active in scope.
TPO (Third Party Organization)	Represents external partners or integrated service providers; currently inactive.
Utility	The power supplier or grid service associated with each location for delivering electricity.
Pricing Policy	A configuration defining how charging fees (energy, session, or flat) are structured and applied.
Payout Account	A financial account used to receive payments, process commissions, or manage disbursements.
Coordination Node ID	An identifier used for energy management and node-level coordination between the grid and charging infrastructure.